PD240W

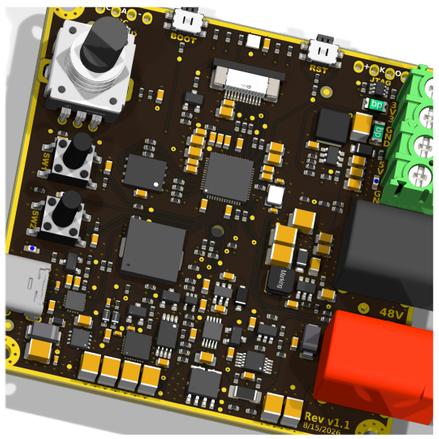
Variant:

2026-10-28

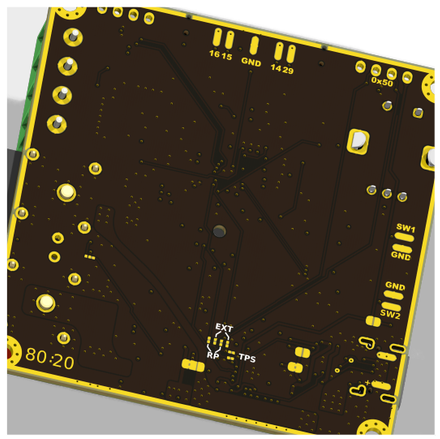
Rev 1.1.0

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Power - Sequencing	21	.....	31	.....
2	Block Diagram	12	Revision History	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	Power - USB-C Input	14	.....	24	.....	34	.....
5	Power - PD Controller	15	.....	25	.....	35	.....
6	Power - Sensing	16	.....	26	.....	36	.....
7	Power - Output	17	.....	27	.....	37	.....
8	MCU - Unit	18	.....	28	.....	38	.....
9	User - Interface	19	.....	29	.....	39	.....
10	Power - Generation	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

An adjustable power supply using USB-C Power Delivery negotiation, supporting up to 240W at 48V 5A. This device is designed to be compatible with USB-PD 3.1 and above.

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 28-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

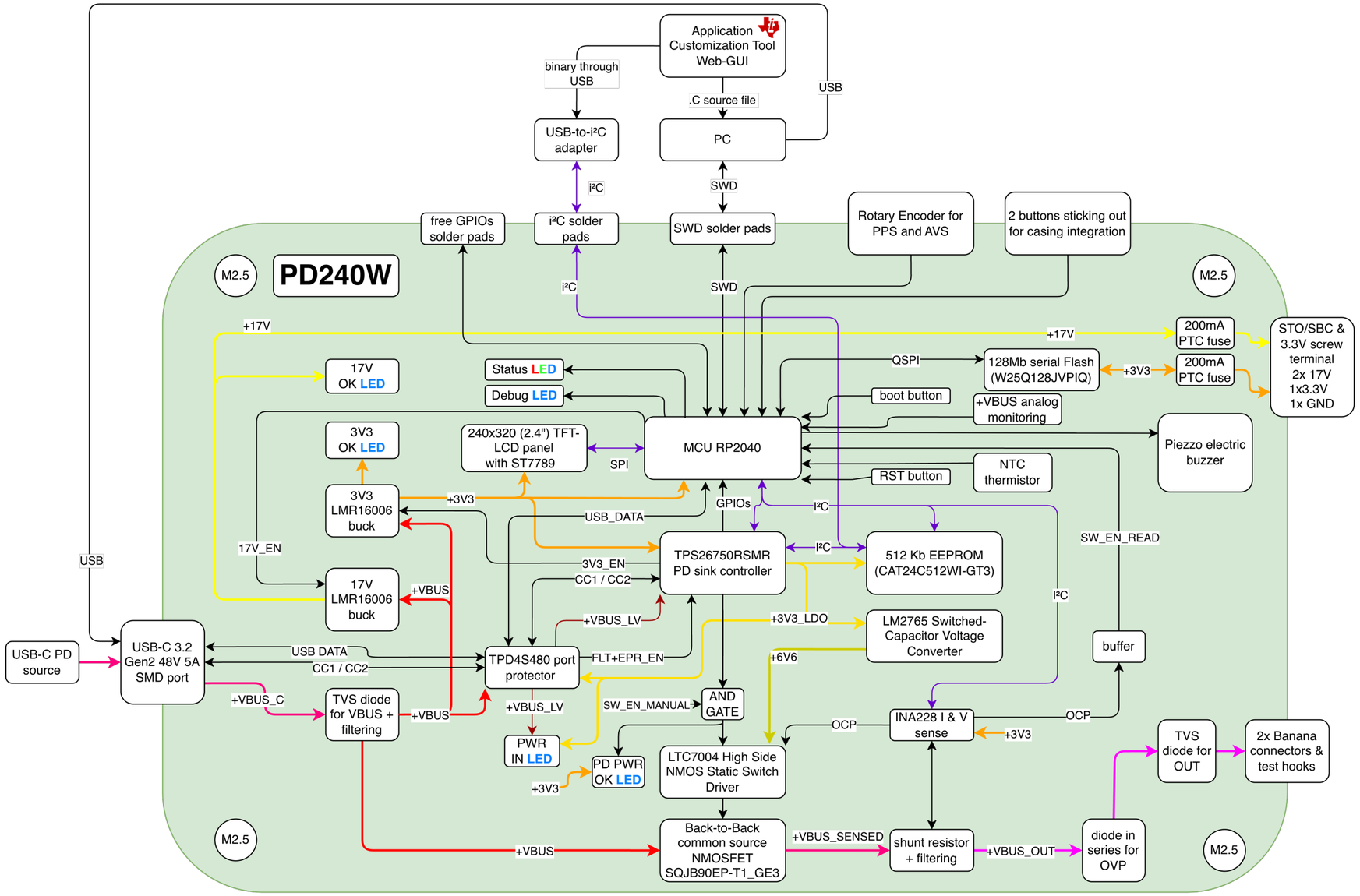
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: **Company Name**		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title:	File Name: PD240W_kicad.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /		Reviewer:	Size: **A3**	Sheet: **1** of **12**

[2] Block Diagram

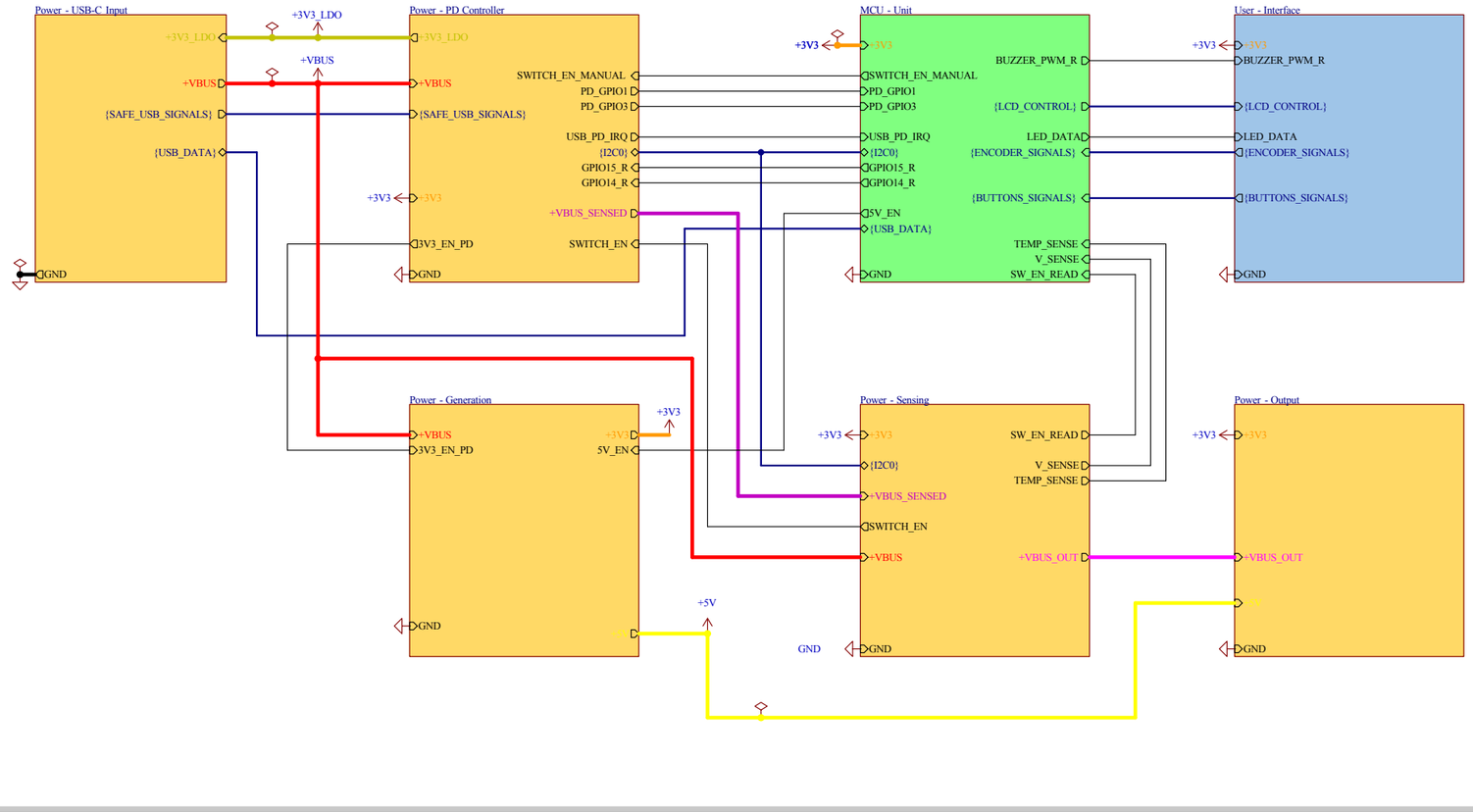


Target specifications:

Input voltage:	5 - 53 V
Output Voltage:	3.3 - 48 V
Max. Output Current:	5 A
Max Power Output:	240 W

Comments:	Company: **Company Name**		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **12**

[3] Project Architecture





Mounting Holes M2.5



Casing Centering Hole



Fiducials - Top

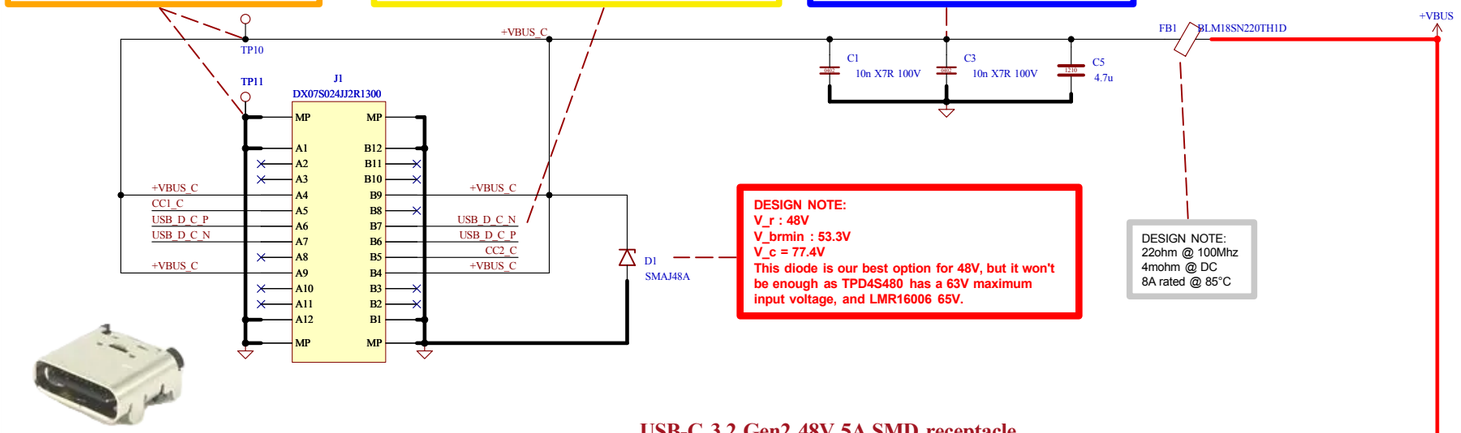
Comments:	Company: **Company Name**		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **12**

[4] Power - USB-C Input

DEBUG NOTE:
+VBUS_C and GND solder pads to test PCB with external supply at limited current.

DESIGN NOTE:
Pin A6 & B6 as well as A7 & B7 need to be connected together, to be able to use he USB-C connector in both directions.

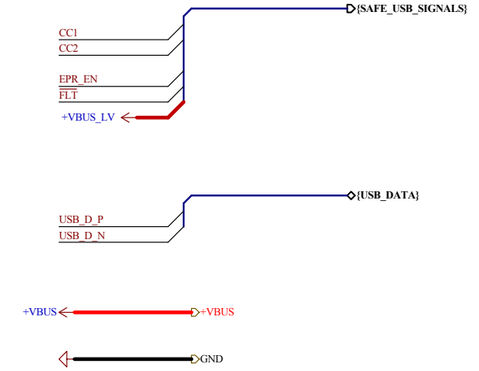
LAYOUT NOTE:
Place the capacitors next to the VBUS pins.



DESIGN NOTE:
V_r : 48V
V_brmin : 53.3V
V_c = 77.4V
This diode is our best option for 48V, but it won't be enough as TPD4S480 has a 63V maximum input voltage, and LMR16006 65V.

DESIGN NOTE:
22ohm @ 100Mhz
4mohm @ DC
8A rated @ 85°C

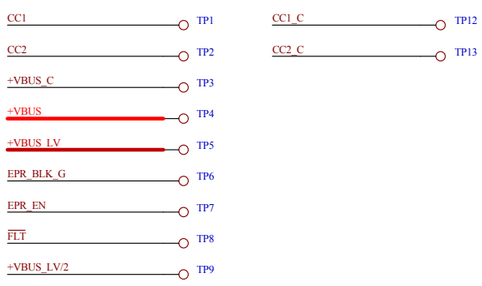
USB-C 3.2 Gen2 48V 5A SMD receptacle



Output Signals



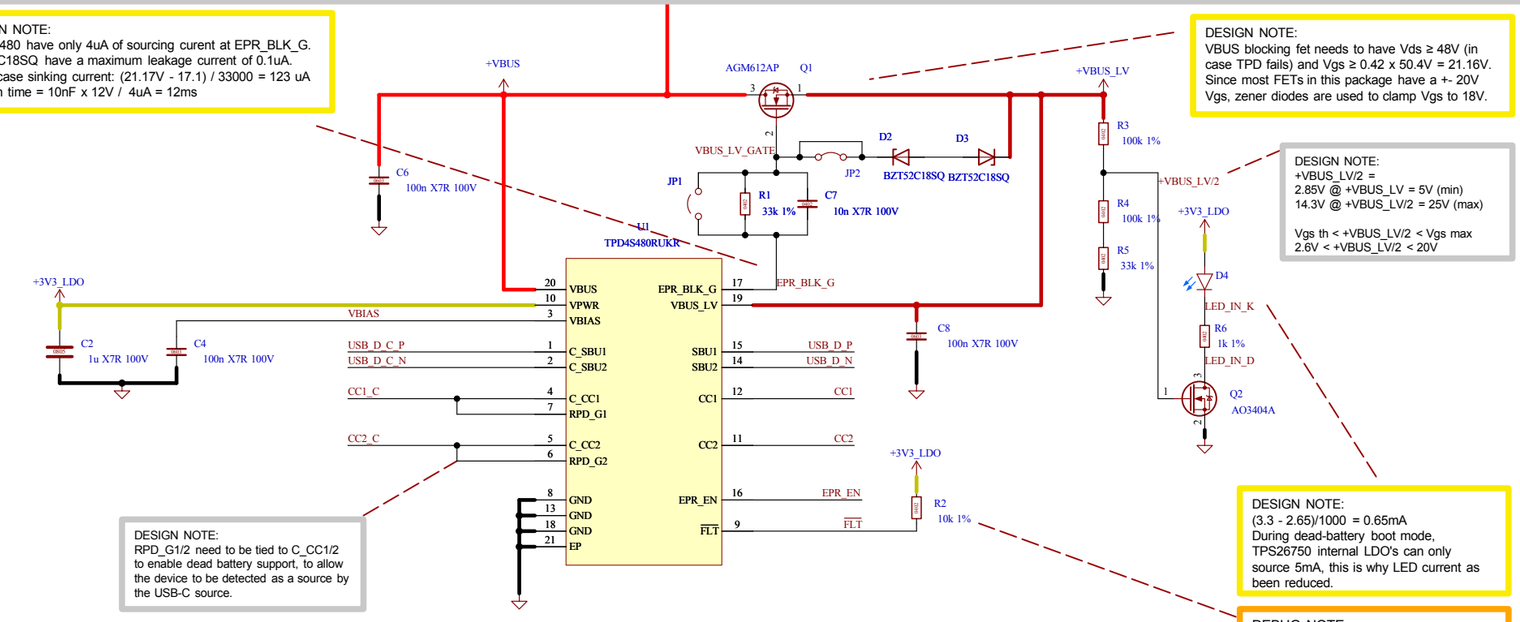
Input Signals



Test Points

DESIGN NOTE:
TPD4S480 have only 4uA of sourcing curent at EPR_BLK_G.
BZT52C18SQ have a maximum leakage current of 0.1uA.
Worst case sinking current: (21.17V - 17.1) / 33000 = 123 uA
Turn-on time = 10nF x 12V / 4uA = 12ms

DESIGN NOTE:
VBUS blocking fet needs to have Vds ≥ 48V (in case TPD fails) and Vgs ≥ 0.42 x 50.4V = 21.16V. Since most FETs in this package have a +- 20V Vgs, zener diodes are used to clamp Vgs to 18V.



DESIGN NOTE:
+VBUS_LV/2 =
2.85V @ +VBUS_LV = 5V (min)
14.3V @ +VBUS_LV/2 = 25V (max)

Vgs th < +VBUS_LV/2 < Vgs max
2.6V < +VBUS_LV/2 < 20V

DESIGN NOTE:
RPD_G1/2 need to be tied to C_CC1/2 to enable dead battery support, to allow the device to be detected as a source by the USB-C source.

DESIGN NOTE:
(3.3 - 2.65)/1000 = 0.65mA
During dead-battery boot mode, TPS26750 internal LDO's can only source 5mA, this is why LED current as been reduced.

DEBUG NOTE:
FLT is open-drain and requires a pull-up. To save board space, it could be removed, as TPS26750 has configurable internal pull-ups.

USB Type-C™ 48V EPR port protector with short-to-VBUS overvoltage and IEC ESD protection

Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Power - USB-C Input	File Name: power_usb_c_input.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/Power - USB-C Input/		Reviewer:	Size: A3	Sheet: 4 of 12

[5] Power - PD Controller

DESIGN NOTE:
The default state for a USB-C port (before or during negotiation, or if negotiation fails) is effectively USB 2.0 specs: 5V @ 500mA. Therefore we want to limit the inrush current. Synapticon drives have an averager of 150uF of input capacitance. As explained in application informations of LTC7004, the gate capacitor and TGUP resistors are chosen to limit the current to a safe limit of arround 250mA.

I_inrush = (0.7x12Vx150uF)/(50kOhm x 100nF) = 252mA

Time to reach 48V = (150uF x 48) / 0.252A = 29ms

Boost capacitor adjustments: SQJB90EP has a max gate charge of 2x25nC = 50nC
The relationship for CB that needs to be maintained when CG is used is given by: Cb > (mosfet Qg)/1V + 10 x CG = 1.05 uF
With derating, a 2.2uF cap is necessary.

DESIGN NOTE:
3V3 LDO total current needs to be lower than 5mA at startup:
EEPROM read: 1mA
POWER OK LED: 0.65mA
TPD4S480: 0.16mA
LM2765: 0.13mA
LTC7004: 0.05mA
I²C pull-ups: ~~0.5mA
ADC resistors: 0.02mA
AND gate: <0.01mA
Total ~~2.5mA < 5mA

DEBUG NOTE:
If the gate driver is not working as expected, connect the bootstrap capacitor to +VBUS instead of TS.

LAYOUT NOTE:
CC1/CC2 capacitors need to be placed as close as possible to their respective pins and on the same layer as the TPS26750 device.

DEBUG NOTE:
ADCIN both conected to LDO1.5V result in ADCINx decoded value 5, which activate AlwaysEnableSink mode and I²C address 0x21 (7 bits). (see page 34)

DESIGN NOTE:
LED current:
(3.3 - 2.65)/330 = 2mA

USB Type-C® and USB PD Controller With Integrated Power Switches

Input Signals

512 Kb I2C CMOS Serial EEPROM

DEBUG NOTE:
When flashing the binary configuration to the EEPROM leave the jumper open, and power the EEPROM from an external supply, to avoid I²C bus conflict with TPS26750.

DEBUG NOTE:
EEPROM Address: Must be 0x50 (7-bit) on the I2Cc bus for the TPS26750 to find it automatically. (1010000)

Output Signals

Test Points

Comments:		Company:		Variant:	Git Hash: 0821683
		Board Name: PD240W		Project Name: PD240W	
Sheet Title: Power - PD Controller		File Name: power_pd_controller.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/Power - PD Controller/			Reviewer:	Size: A3	Sheet: 5 of 12

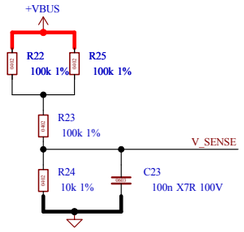
[6] Power - Sensing



Input Signals

DESIGN NOTE:
Network ratio = 1/16 = 0.0625
@48V V_SENSE = 3V
@5V V_SENSE = 0.31V

LAYOUT NOTES:
Place capacitor close to MCU DC pin. Route away from noisy traces.

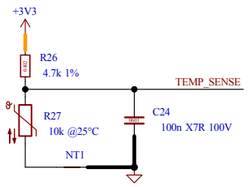


Input Voltage Sensing

DESIGN NOTE:
B-Constant is 3950K
1.2k @ 80°C -> 0.67V
2.5k @ 60°C -> 1.14V
10k @ 25°C -> 2.24V
13k @ 20°C -> 2.42V

LAYOUT NOTES:
NTC must be placed close and thermally coupled to the ground plane near the relevant heat source. Route away from noisy traces.

DEBUG NOTE:
Low side measurement is used to improve thermal coupling, which invert the maths in software.

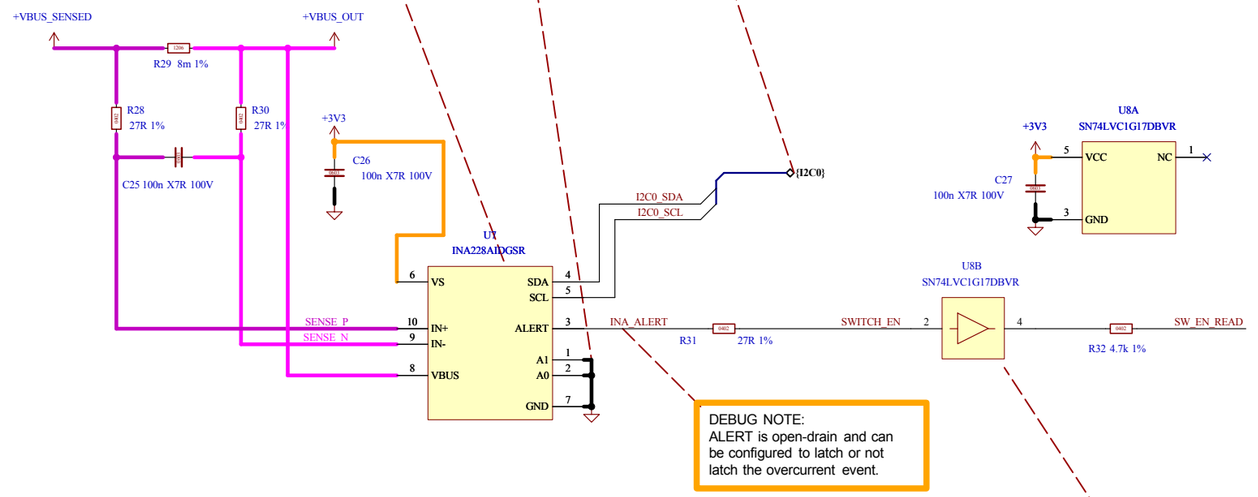


Temperature Sensing

DESIGN NOTE:
ADCRANGE = 1 (+- 40.96 mV range)
V_{SENSE} = 5A x 0.008 = 40 mV. This utilizes 97.6% of the ADC range, providing the highest possible resolution.
Power Dissipation: P = 5² x 0.008 = 0.2W

DEBUG NOTE:
INA228AIDGSR address when A1 & A0 are tied to GND is 0x40 (7 bits 1000000) (see page 19)

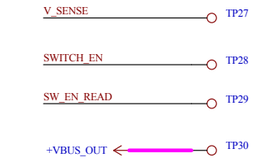
DESIGN NOTE:
I2C pullups are on Power - PD Conroller schematic



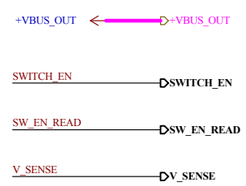
DEBUG NOTE:
ALERT is open-drain and can be configured to latch or not latch the overcurrent event.

DESIGN NOTE:
SN74LVC1G17 is a safety buffer to prevent software mistakes. Driving SWEN_READ cannot activate the power switch.

85-V, 20-Bit, Ultra-Precise Power/Energy/Charge Monitor With I²C Interface



Test Points



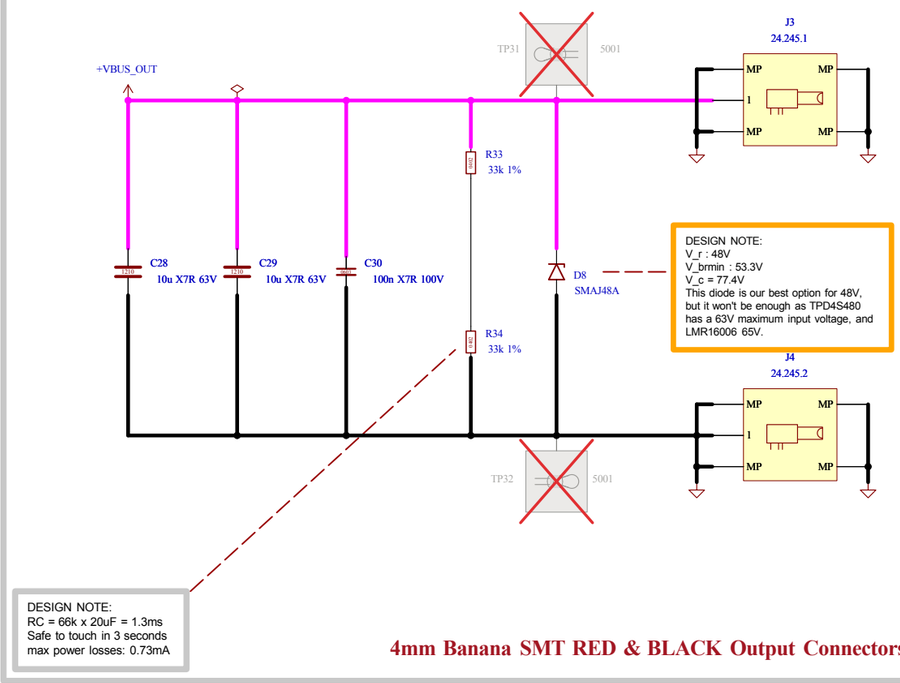
Output Signals

Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Power - Sensing	File Name: power_sensing.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/Power - Sensing/		Reviewer:	Size: **A3**	Sheet: **6** of **12**

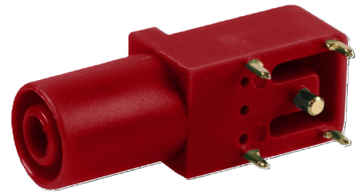
[7] Power - Output



Input Signals

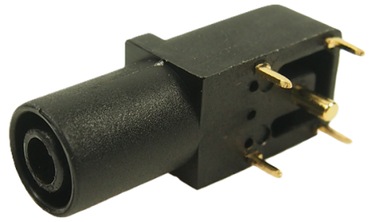


4mm Banana SMT RED & BLACK Output Connectors

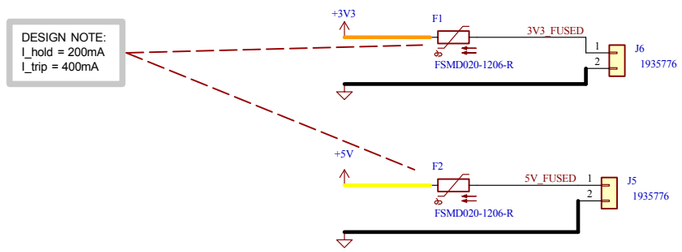


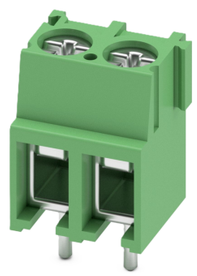
FCR7350R - C7437326

LAYOUT NOTES:
Space Banana connectors by a standard 19.05mm.
Add mechanical stitching vias to +VBUS_OUT and GND to provide mechanical anchors against external torque applied to the connectors.



FCR7350B - C7437327



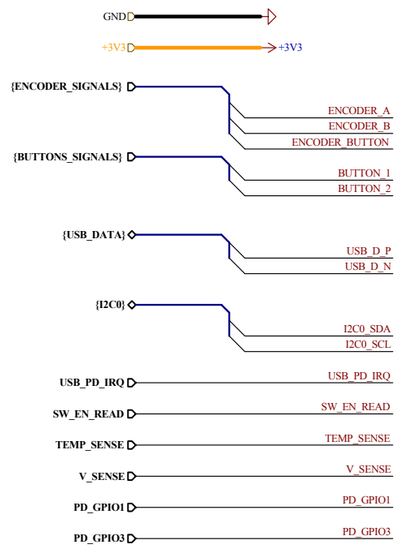


Screw Terminals for 5V

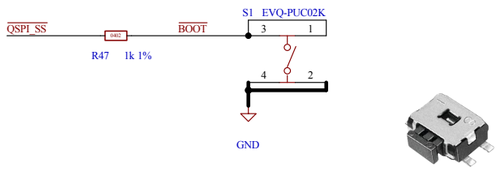
Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Power - Output	File Name: power_output.kicad_sch	Designer: Theo Heng	Date: 2026-08-15	Revision: 1.1.0
Sheet Path: /Project Architecture/Power - Output/		Reviewer:	Size: **A3**	Sheet: **7** of **12**

[8] MCU - Unit

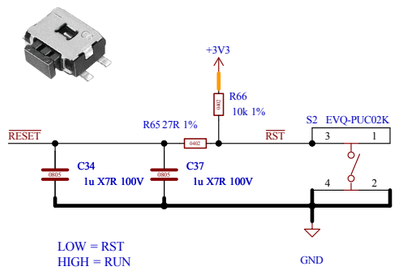
Input Signals



MCU Boot Button



MCU Reset Button



DESIGN NOTE:
We want to delay the start of the MCU to allow power to stabilize and avoid flash corruption. RP2040 is assured to be turned on when the RUN pin reaches 2.0V, but can tun on at 0.8V already. Run has an internal pull-up of 50k-80k. ->Effective pull-up is between 8.3k and 8.8k

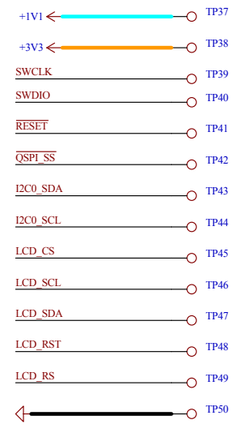
RCmin = 8.3k x 2uF = 16.6ms
RCmax = 8.8k x 2uF = 17.6ms

Worst case (fastest turn-on):
V(t) = V x (1-e^(-t/RC))
0.8 = 3.3 x (1-e^(-t/RC))
-t/RC = ln(0.757)
t = 0.28 x RCmin = 4.6ms

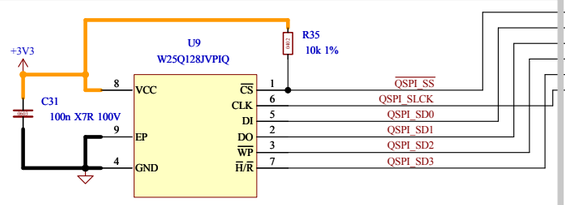
Has been verified in Falstad simulation.

Worst case (slowest turn-on):
2 = 3.3 x (1-e^(-t/RC))
-t/RC = ln(0.394)
t = 0.931 x RCmax = 16.38ms

Test Points

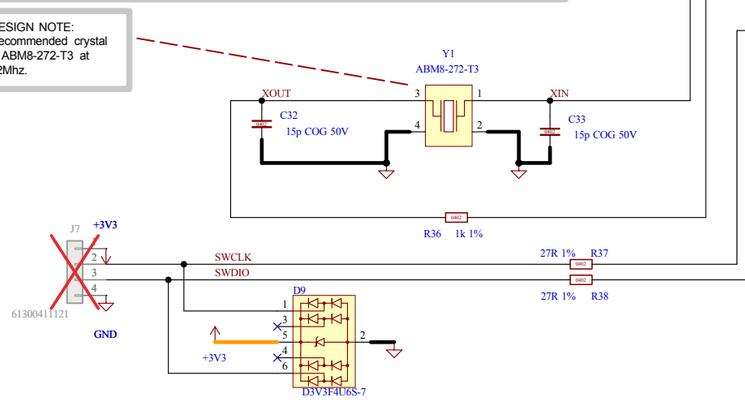


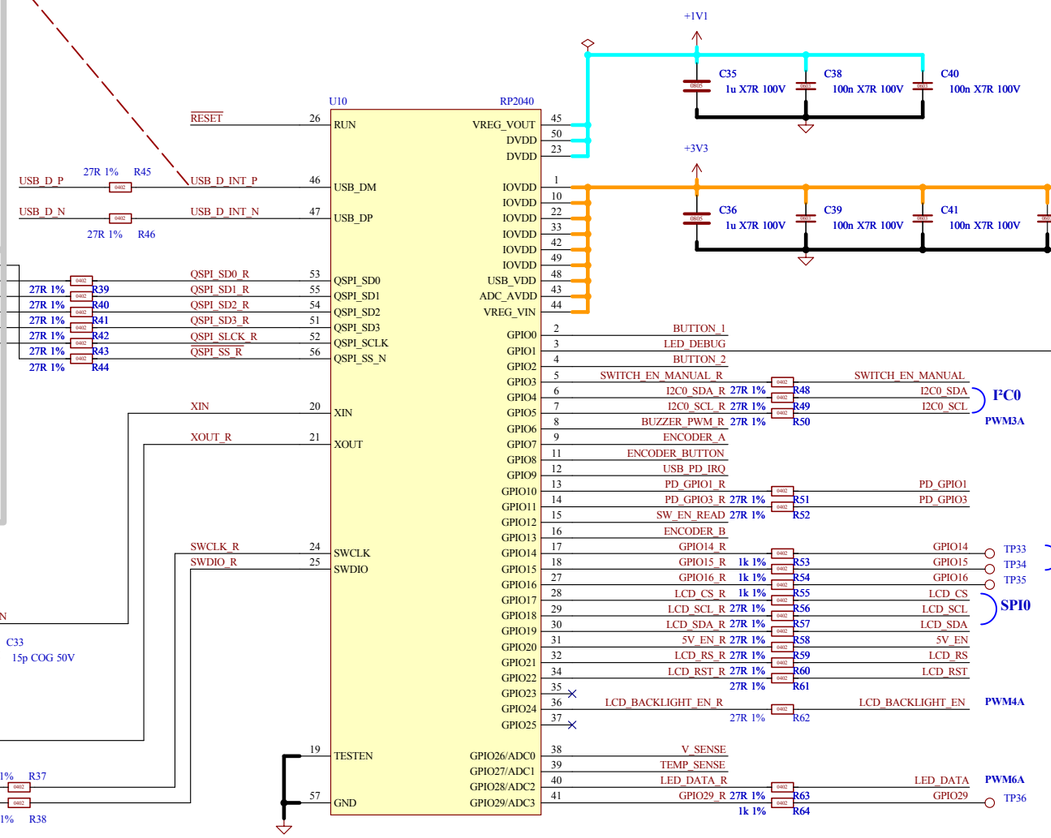
128Mb Serial Flash Memory

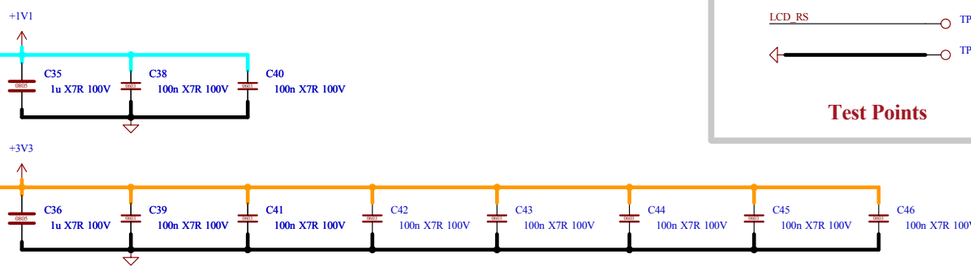


LAYOUT NOTES:
D+ and D- lines of USB-C needs to be roued as differential pairs, with 90Ω impedance.
Spacing: 0.2032mm
Width: 0.15mm

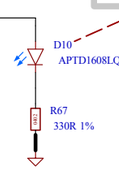
DESIGN NOTE:
Recommended crystal is ABM8-272-T3 at 12Mhz.



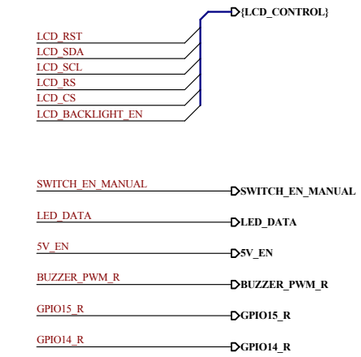




DESIGN NOTE:
LED current: (3.3 - 2.65) / 330 = 2mA



Output Signals



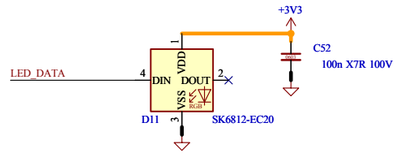
MCU - Unit

Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: MCU - Unit	File Name: mcu_unit.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/MCU - Unit/		Reviewer:	Size: **A3**	Sheet: **8** of **12**

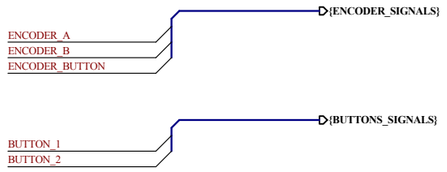
[9] User - Interface



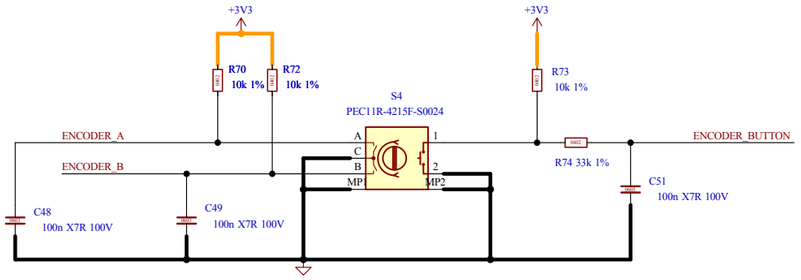
9x9mm SMD 65dB piezoelectric buzzer



Addressable RGD LED



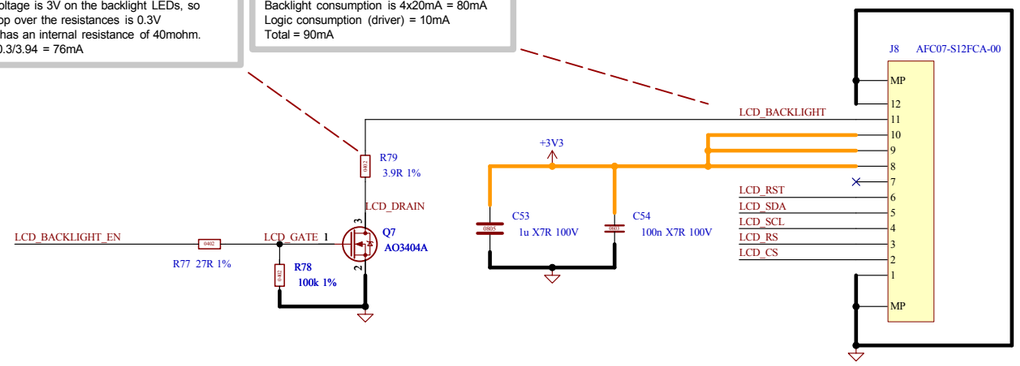
Output Signals



24 PPR Rotary Encoder with 24 Detents & Button

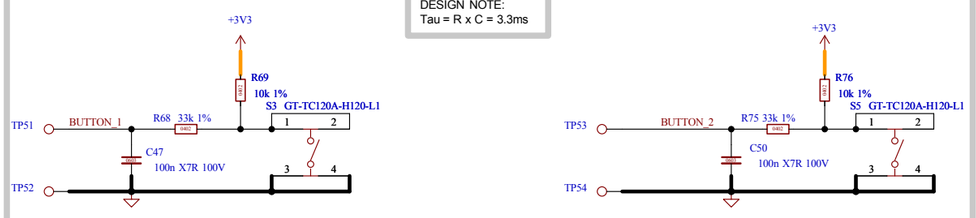
DESIGN NOTE:
Forward voltage is 3V on the backlight LEDs, so voltage drop over the resistances is 0.3V
AO3404A has an internal resistance of 40mohm.
current = 0.3/3.94 = 76mA

DESIGN NOTE:
Backlight consumption is 4x20mA = 80mA
Logic consumption (driver) = 10mA
Total = 90mA





12-pins, Low-side Contact FPC Connector and Backlight Driver



2x User Tactile Switches

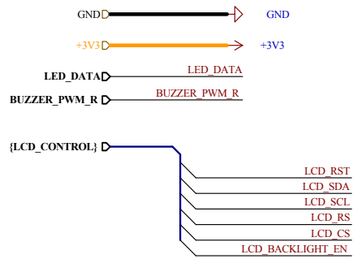
LAYOUT NOTES:
FPC connector of LCD display is 20.7mm long, which allows for arround 10mm of clearance between the top of the PCB and the bottom of the display.

DISPLAY DSP1 HS20HS072RX

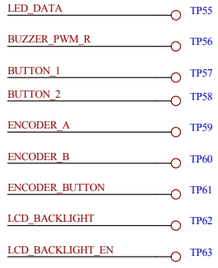
NO.	SYMBOL
1	GND
2	CS
3	RS
4	SCL
5	SDA
6	RESET
7	NC
8	IOVCC
9	VCC
10	LEDA
11	LEDK
12	GND



240x320 (2.4") TFT-LCD Panel with ST7789 driver & FPC

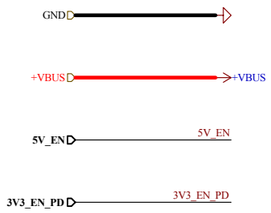


Input Signals

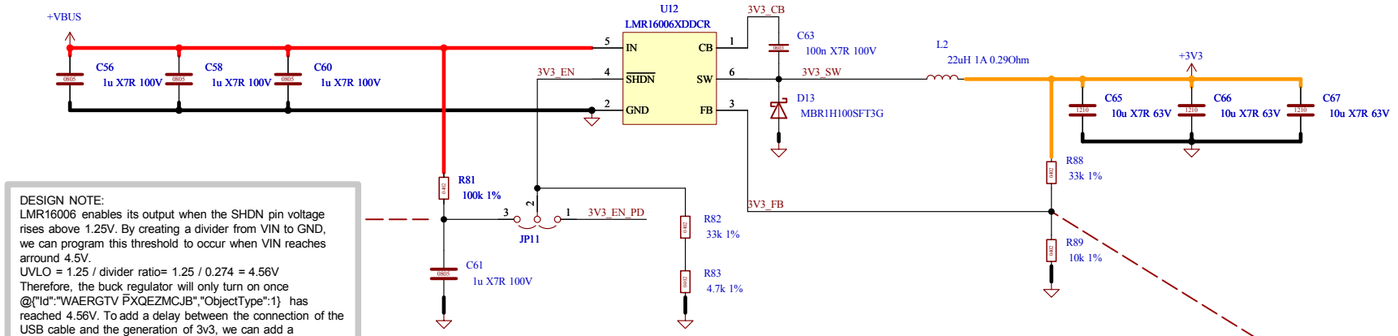


Test Points

Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: PD240W		Project Name: PD240W	
Sheet Title: User - Interface	File Name: user_interface.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/User - Interface/		Reviewer:	Size: A3	Sheet: 9 of 12



Input Signals



DESIGN NOTE:
LMR16006 enables its output when the SHDN pin voltage rises above 1.25V. By creating a divider from VIN to GND, we can program this threshold to occur when VIN reaches arround 4.5V.
UVLO = 1.25 / divider ratio= 1.25 / 0.274 = 4.56V
Therefore, the buck regulator will only turn on once @{"Id":"WAERGTV P̄XQEZMCJB","ObjectType":1} has reached 4.56V. To add a delay between the connection of the USB cable and the generation of 3v3, we can add a capacitor.
From Thevenin equivalent resistance:
RC = 27k x 1uF = 27ms

V(t) = Vfinal x (1-e^(-t/RC))
1.25= 1.37 x (1-e^(-t/RC))
-t/RC = ln(0.087)
t = 2.44 x RC = 66ms
LMR16006 should turn on 66ms after VBUS has reached 5V.

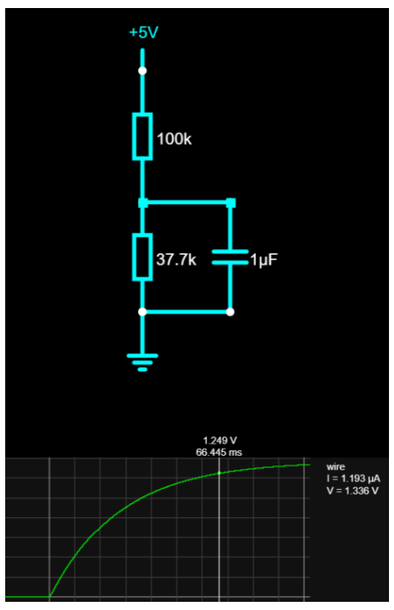
3.3V 600mA 700kHz Buck Regulator

DESIGN NOTE:
FB: Vout = 0.765 x (1 + Rtop/Rbot)
= 0.765 x (1 + (33k / 10k) = 2.29V

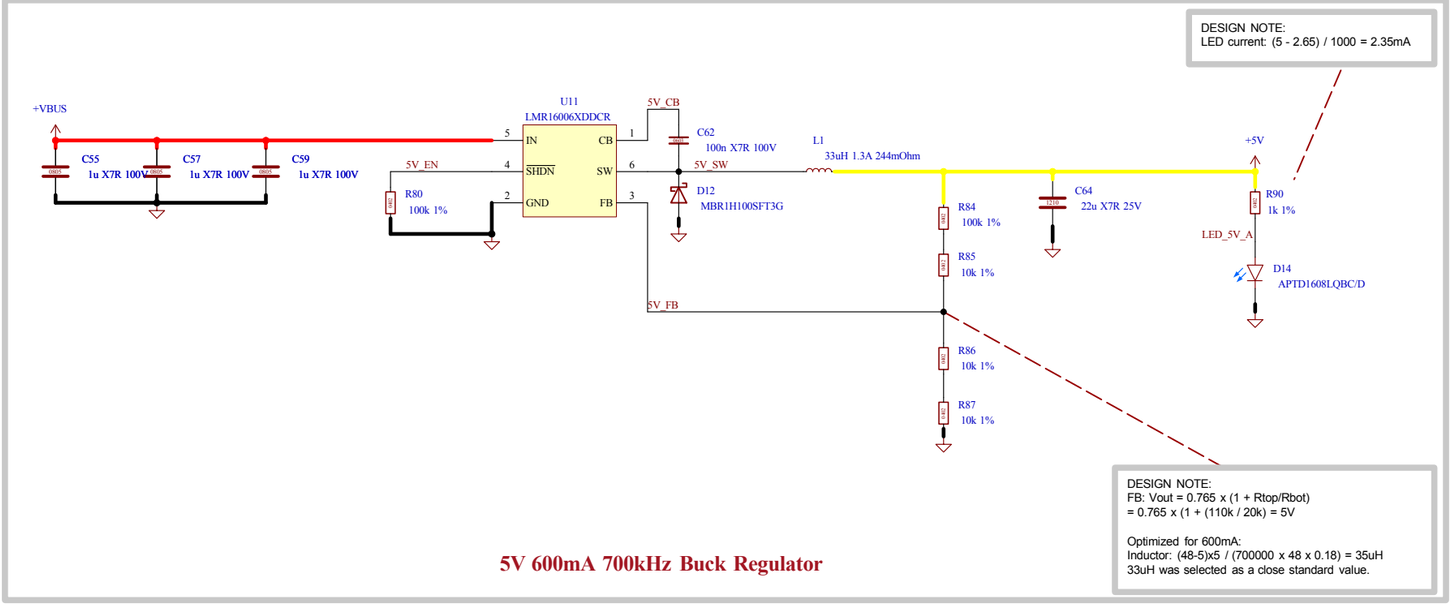
Optimized for 600mA:
Inductor: (48-3.3)x3.3 / (700000 x 48 x 0.18) = 24uH
22uH was selected as a close standard value.



Output Signals



Falstad Simulation Results: 1.25V reached in 66ms.

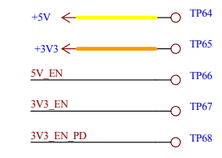


DESIGN NOTE:
LED current: (5 - 2.65) / 1000 = 2.35mA

DESIGN NOTE:
FB: Vout = 0.765 x (1 + Rtop/Rbot)
= 0.765 x (1 + (110k / 20k) = 5V

Optimized for 600mA:
Inductor: (48-5)x5 / (700000 x 48 x 0.18) = 35uH
33uH was selected as a close standard value.

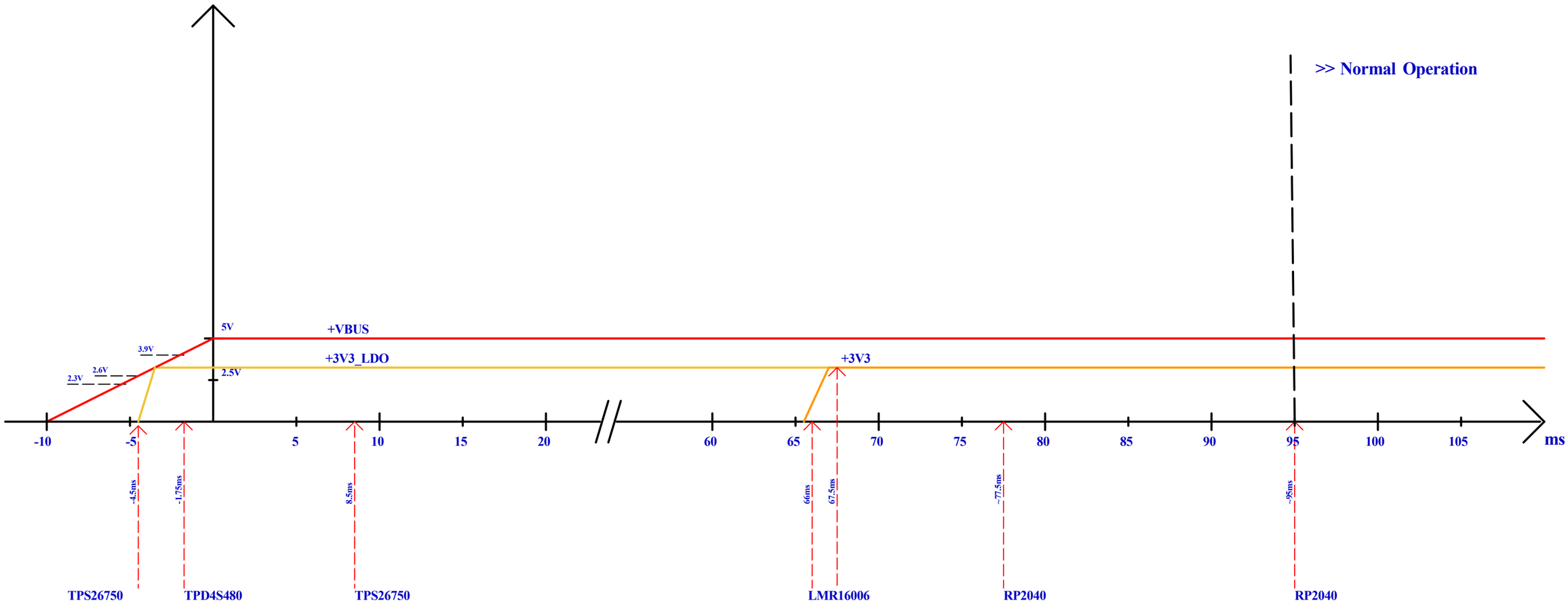
5V 600mA 700kHz Buck Regulator



Test Points

Comments:	Company:		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Power - Generation	File Name: power_generation.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: **A3**	Sheet: **10** of **12**

[11] Power - Sequencing



Comments:	Company: **Company Name**		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Power - Sequencing/		Reviewer:	Size: **A3**	Sheet: **11** of **12**

[12] Revision History

Version 1.0.0 - 2026-03-15

Added

- Original release by Théo Heng (theohg/PD240W)

Version 1.1.0 - 2026-08-15

Changed

- Auxiliary buck regulator output changed from 17V to 5V: new feedback divider, 33µH inductor, 22µF 25V output capacitor
- JLCPCB production files and footprint rotation offsets added

Known Issues

- USB D+/D- lines are still swapped and need the bodge described in the README

Comments:	Company: **Company Name**		Variant:	Git Hash: 0821683
	Board Name: **PD240W**		Project Name: **PD240W**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Theo Heng	Date: 2026-02-27	Revision: 1.1.0
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **12** of **12**